

First step Analysis.

$$\bigcup \left\{ X_n \in C_j \right. \\ \left. \text{for some } n \geq 1 \right\}$$

$$S = \underbrace{C_1} \cup \underbrace{C_2} \cup \dots \cup \underbrace{T}$$

rec.

transient states.

If T is finite, then for $i \in T$

$$P(X_n \in T \text{ for all } n \geq 1 \mid X_0 = i) = 0.$$

$$\Rightarrow P(X_n \in \bigcup C_j \mid X_0 = i) = 1 \Rightarrow$$

for some $n \geq 1$

$$\Rightarrow \sum_j P(X_n \in C_j \text{ for some } n \geq 1 | X_0 = i) = 1.$$

Defn :- Eventual absorption in rec. class C_j is defined as

$P(X_n \in C_j \text{ for some } n \geq 0 | X_0 = i)$
 Gambler's Ruin problem: $C_1 = \{0\}, C_2 = \{N\},$
 $T = \{1, 2, \dots, N-1\}.$

For $i \in T$, $P(X_n \in C \text{ for some } n \geq 0 | X_0 = i)$.

= Prob of loss starting
with initial capital i .

Let C be a closed class.

Define $\alpha_i^{(C)} = P(X_n \in C \text{ for some } n \geq 0 | X_0 = i)$
 $= \begin{cases} 0 & \text{if } i \in D, C \cap D = \emptyset, D \text{ is closed} \\ 1 & \text{if } i \in C \end{cases}$

Define $\alpha_i^{(c)}(n) = P(X_n \in C \mid X_0 = i)$
 for $i \in T$.

$A_n = \{X_n \in C\}$, $A_{n+1} = \{X_{n+1} \in C\}$.

$A_n \subseteq A_{n+1}$.

$$\bigcup_{n=0}^{\infty} A_n = \{X_n \in C \text{ for some } n \geq 0 \mid X_0 = i\}$$

$$\alpha_i^{(c)} = P\left(\bigcup_{n=0}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \alpha_i^{(c)}(n)$$

$$\alpha_i^{(c)} = \lim_{n \rightarrow \infty} \alpha_i^{(c)}(n)$$

$$\alpha_i^{(c)}(0) = 0.$$

$$\alpha_i^{(c)}(1) = \sum_{j \in C} p_{ij} = p_{i,c}.$$

$$\begin{aligned} \alpha_i^{(c)}(n+1) &= P(X_{n+1} \in C \mid X_0 = i) \\ &= \sum_{j \in C} P(X_1 = j, X_{n+1} \in C \mid X_0 = i) \end{aligned}$$

$$\begin{aligned} &+ \sum_{j \in T} P(\dots) \\ &+ \sum_{j \notin T \cup C} P(\dots) \\ &0 \end{aligned}$$

$$= \sum_{j \in C} p_{ij} + \sum_{j \in T} P(X_1=j | X_0=i) P(X_{n+1} \in C | X_1=j, X_0=i)$$

$$= p_{i,C} + \sum_{j \in T} p_{ij} \underbrace{P(X_n \in C | X_0=i)}_{\alpha_j^{(n)}}$$

$$\alpha_i^{(n+1)} = p_{i,C} + \sum_{j \in T} p_{ij} \alpha_j^{(n)}$$

Taking limit as $n \rightarrow \infty$, $\alpha_i^{(\infty)} = p_{i,C} + \sum_{j \in T} p_{ij} \alpha_j^{(\infty)}$

Let $\{y_i : i \in T\}$ be a solⁿ of

the system of eqⁿ s.t.

$$0 \leq y_i \leq 1. \\ \forall i \in T$$

$$y_i = p_{i,c} + \sum_{j \in T} p_{ij} y_j$$

For any $i \in T$,

$$y_i \geq 0 = \alpha_i^{(0)}(0)$$

Want to show

$$y_i \geq \alpha_i^{(n)}(u)$$

$$0 \leq u \leq 1$$

Assume the result for n , i.e., $y_i \geq \alpha_i^{(n)}(n)$
 $\forall i \in T$.

$$\begin{aligned} y_i &= p_{i,c} + \sum_{j \in T} p_{ij} y_j \\ &\geq p_{i,c} + \sum_{j \in T} p_{ij} \alpha_j^{(n)}(n) \\ &= \alpha_i^{(n)}(n+1). \end{aligned}$$

Thm :- The eventual absorption prob to a closed class C is given by the

Smallest soln of the system of linear eqn.

$$y_i = p_{i,c} + \sum_{j \in T} p_{ij} y_j$$

with

$$y_i \in [0, 1] \quad \forall \underline{i} \in T.$$

Ex 1. -

$$C_1 = \{0, 3\}, C_2 = \{2, 4, 7\}, T = \{1, 6, 5\}.$$

$\alpha_1^{(c_1)}, \alpha_6^{(c_1)}, \alpha_5^{(c_1)}$

$$y_1 = \frac{1}{2} + p_{11}y_1 + p_{15}y_5 + p_{16}y_6$$

$$y_5 = \frac{1}{4} + p_{51}y_1 + p_{55}y_5 + p_{56}y_6$$

$$y_6 = \frac{1}{3} + p_{61}y_1 + p_{65}y_5 + p_{66}y_6$$

$$p_{1,C_1} = \frac{1}{2}$$

$$p_{5,C_1} = \frac{1}{4}$$

$$p_{6,C_1} = \frac{1}{3}$$

Ex: Birth & Death chain

$$P = \begin{array}{c|cccc} & 0 & 1 & \dots & N \\ \hline 0 & r_0 & p_0 & & 0 \\ 1 & q_1 & r_1 & p_1 & \\ 2 & 0 & q_2 & r_2 & p_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ N-1 & 0 & & q_{N-1} & r_{N-1} & p_{N-1} \\ N & & & & q_N & r_N \end{array}$$

$$\alpha_i^{(C_1)} = P(X_n \in C_1 \text{ for some } n \geq 0 \mid X_0 = i), \quad i \in T.$$

$$p_i, q_i \geq 0 \text{ for } i=1, \dots, N-1$$

$$r_0 = r_N = 1.$$

$$\Rightarrow p_0 = p_N = 0$$

$$C_1 = \{0\}, C_2 = \{N\}.$$

$$T = \{1, 2, \dots, N-1\}$$

$$y_1 = a_1 + r_1 y_1 + b_1 y_2 = a_1 y_0 + r_1 y_1 + b_1 y_2$$

$$y_i = a_i y_{i-1} + r_i y_i + b_i y_{i+1}, \quad i=2, \dots, N-2,$$

$$y_{N-1} = a_{N-1} y_{N-2} + r_{N-1} y_{N-1} = a_{N-1} y_{N-2} + r_{N-1} y_{N-1}$$

Set $y_0 = 1, y_N = 0$, then $+ b_{N-1} y_N$

$$y_i = a_i y_{i-1} + r_i y_i + b_i y_{i+1}, \quad i=1, \dots, N-1$$

$$\Rightarrow (q_i + p_i) y_i = q_i y_{i-1} + p_i y_{i+1}$$

$$\Rightarrow q_i (y_i - y_{i-1}) = p_i (y_{i+1} - y_i)$$

$$\Rightarrow y_{i+1} - y_i = \frac{q_i}{p_i} (y_i - y_{i-1})$$

$$= \prod_{j=1}^i \frac{q_j}{p_j} (y_1 - y_0) = \frac{q_i}{p_i} \cdot \frac{q_{i-1}}{p_{i-1}} (y_{i-1} - y_{i-2}) \quad \text{for } i=1, \dots, N-1.$$

$$y_{i+1} - y_i = \prod_{j=1}^i \frac{a_j}{p_j} (y_1 - y_0), \quad i=1, \dots, N-1.$$

$$\text{Set } a_i = \begin{cases} 1 & \text{if } i=0 \\ \prod_{j=1}^i \frac{a_j}{p_j} & \text{if } i > 0 \end{cases} \quad \underline{= a_i (y_1 - y_0)}$$

$$\begin{aligned} y_N - y_0 &= \sum_{i=0}^{N-1} (y_{i+1} - y_i) \\ &= \sum_{i=0}^{N-1} a_i [y_1 - y_0] \\ &= (y_1 - y_0) \sum_{i=0}^{N-1} a_i \end{aligned}$$

$$\Rightarrow y_1 - y_0 = \frac{1}{\sum_{i=0}^{N-1} a_i}$$

$$\left[\sum_{i=0}^{\infty} a_i = \infty \right] \Rightarrow \text{Eventual absorption at 1 w.p. 1}$$

$$y_{k-1} = y_k - y_0 = \sum_{i=0}^{k-1} (y_{i+1} - y_i) = \sum_{i=0}^{k-1} a_i (y_1 - y_0)$$

$$\Rightarrow y_k = 1 - \frac{\sum_{i=0}^{k-1} a_i}{\sum_{i=0}^{N-1} a_i}$$

$$= \sum_{i=0}^{k-1} a_i \times \left(\frac{1}{\sum_{i=0}^{N-1} a_i} \right)$$

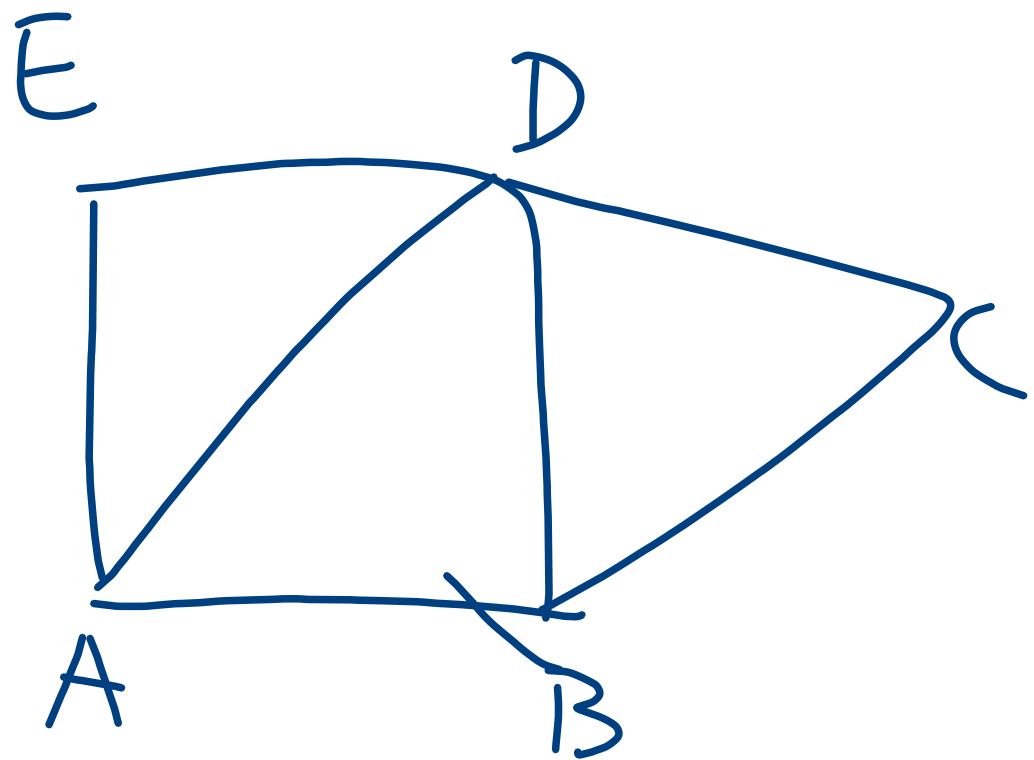
$$y_k = 1 - \frac{\sum_{i=0}^{k-1} a_i}{\sum_{i=0}^{N-1} a_i} \quad i=1, \dots, N-1$$

Gambler's ruin prob :- $p_i = p, q_i = q$.

$$\Rightarrow a_i = \prod_{j=1}^i \frac{q}{p} = \left(\frac{q}{p}\right)^i$$

if $p = q = 1/2$, $a_i = 1 \Rightarrow y_k = 1 - \frac{k}{N}$.

In the infinite case, if $q \geq p$, Eventual ruin happens w.p. 1. otherwise, Eventual ruin prob < 1 .



$$P(X_n = C \text{ for some } n \geq 1, X_t \neq D \text{ for } t < n \mid X_0 = A)$$

$$\tilde{P} = ((\tilde{P}_{ij})) \quad \text{where}$$

$$\tilde{P}_{ij} = \begin{cases} P_{ij}, & \text{if } i \notin \{C, D\} \\ 1, & \text{if } i=j, i \in \{C, D\} \\ 0, & \text{if } i \neq j, i \in \{C, D\} \end{cases}$$

$$P = \begin{matrix} & \begin{matrix} A & B & C & D & E \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{pmatrix} \checkmark & \checkmark & \checkmark & \checkmark & \checkmark \\ \checkmark & \checkmark & \checkmark & \checkmark & \checkmark \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \checkmark & \checkmark & \checkmark & \checkmark & \checkmark \end{pmatrix} \end{matrix}$$

$$\alpha_A^{(1)}, \alpha_B^{(1)}, \alpha_E^{(1)}$$

$$C_1 = \{C\}, C_2 = \{D\}, \\ T = \{A, B, E\}$$

$$y_A = 0 + \frac{1}{3} y_B + \frac{1}{3} y_E$$

$$y_B = \frac{1}{3} + \frac{1}{3} y_A$$

$$y_E = 0 + \frac{1}{2} y_A$$